

ZVS Turn-on Integrated Triangular Current Mode Control for Three-Phase Three-Level Flying Capacitor Bidirectional PFC

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Abstract—As power demand for on-board chargers (OBC) is increasing to meet the fast-charging requirements, 800 V DC bus voltage is being adopted for PFCs to reduce the copper losses in OBCs. For higher voltage operation, multi-level converter topologies employ the benefit of reducing device stress and common-mode voltage. The flying capacitor multi-level (FCML) converter has the added advantage of frequency multiplication. However, switching frequency is still limited by the high turn-on losses, hence ZVS turn-on integrated Triangular Current Mode (iTCM) control is presented in this paper for multi-level PFCs. As compared to Triangular Current Mode (TCM), iTCM has the benefit of very small high frequency current in the main boost inductor which is achieved by adding an LC branch to the converter such that the high frequency current is filtered by this branch. Hence, iTCM is introduced in this paper for three-level FC converter with one single common LC branch for ZVS turn-on of all the devices. The proposed technique is implemented on a GaN based three-level FC 800 V PFC at 11 kW. The efficiency achieved is 98.6% at a minimum inductor current frequency of 330 kHz.

Keywords— *Flying Capacitor Converter, Power Factor Corrector (PFC), Sinusoidal Pulse Width Modulation (SPWM), Triangular Current Mode (TCM), Integrated Triangular Current Mode (iTCM), Continuous Conduction Mode (CCM), Neutral Point Clamped (NPC)*

I. INTRODUCTION

For high power (>10 kW) charging systems, increasing the PFC DC bus voltage to >600 V is imminent owing to the high copper losses at 400 V. Three-phase two-level Si/SiC devices-based converters are the most common choice for 800 V bulk voltage as two-level converters are simple to design and control. However as multi-level converter topologies employ the benefit of reducing device stress and common-mode voltage, they are also a potential candidate for EV chargers operating at 800 V. The most common bidirectional multi-level topologies are Active Neutral Point Clamped (ANPC), Diode Neutral Point Clamped (DNPC), Cascaded H-bridge (CHB), T-type and Flying capacitor converter [1]&[3]. For an N-level converter,

each device will block a voltage of $V_{DC}/(N-1)$. Though T-type topologies don't give the benefit of reducing the blocking voltage for all devices [2]. The flying capacitor converter (shown in Fig.1(b)) has an added advantage of multiplying the inductor current frequency (f_{iL}) by N-1 times the switching frequency (f_{sw}) [3] s.t.

$$f_{iL} = (N-1) * f_{sw} \quad (1)$$

This leads to the benefit of reducing the size of the inductor and EMI filter while keeping the switching losses lower. As compared to SiC based two-level six-switch converters operating at 800 V, GaN based three-level converters have shown to achieve superior performance [6].

However, the achievable power density and efficiency are still limited by the high switching losses at high switching frequencies specifically the turn-on losses. The most effective way to reduce switching losses is by employing soft-switching techniques. Soft-switching can be achieved with TCM [2],[4]&[7]. In [7], TCM combined with Discontinuous Pulse Width Modulation (DPWM) is shown to achieve 98.4% efficiency at 800 V for two-level SiC converter. However, TCM results in very high ripple on the boost inductor thus increasing the current stress on it. Hence, optimizing this inductor becomes challenging as it carries both the high frequency (HF) ripple as well as the low frequency (LF) fundamental current. To overcome this issue, iTCM is introduced in [5] for single phase two-level converters and is expanded to three-phase two-level converters in [8] & [9]. By adding this LC -circuit to the PWM AC/DC converter as shown in Fig. 1(a), TCM current is split into high frequency (HF) and low frequency (LF) components which flow through separate inductors. This principle of operation is extended to three-level FCML converters with just one LC branch (as shown in Fig. 1(b)) reversing the current for both sets of complementary devices in this paper. Furthermore, DPWM requires no connection between AC neutral and DC bus midpoint whereas this connection is essential for OBC applications. Hence, iTCM based on sinusoidal PWM (SPWM) is presented in this paper.

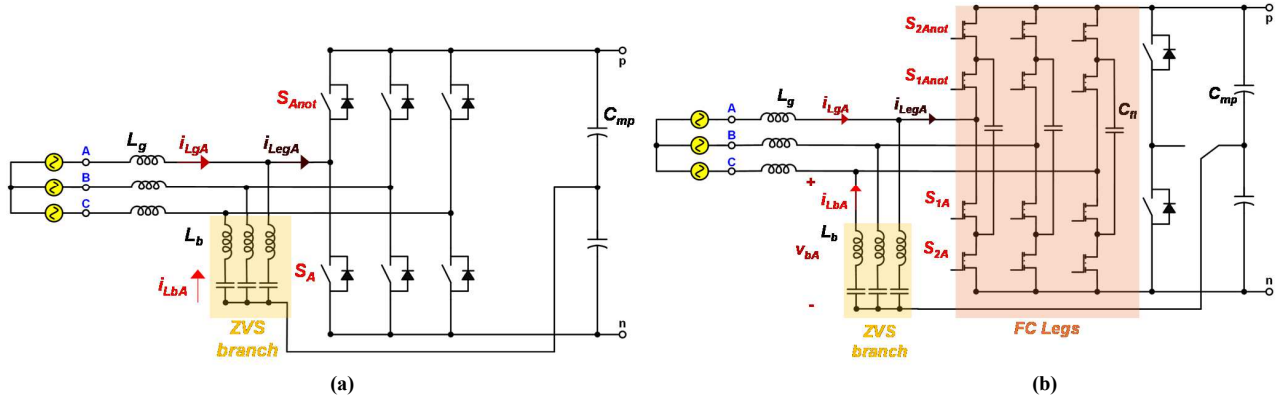


Fig. 1. (a) Three-Phase two-level PFC operating in iTCM mode from [8]. (b) Three-phase/ single phase combo three-level flying capacitor converter with integrated LC branch to achieve soft-switching iTCM operation

The detailed switching operation in three-phase iTCM SPWM and the frequency multiplication effect is discussed in Section II. The parameter design is discussed comprehensively in section III. The detailed design of the converter including GaN based three-level phase leg and inductors is discussed in section IV. The experimental results are discussed in Section V.

II. BASIC OPERATION

For a three-phase converter, when phase neutral is connected to DC bus mid-point (as shown in Fig. 1), each phase can operate as an independent half bridge single-phase converter as shown in Fig. 2(e). For a balanced symmetrical operation, the flying capacitor must be charged to $V_{dc}/2$, i.e. $V_{Cf} = V_{dc}/2$ [6]. The converter operation in CCM is discussed briefly here as iTCM operation is like CCM with variable switching frequency control. The detailed operation in CCM is discussed in [6]. The two pairs of complementary devices, S_{1A} and S_{1Anot} and S_{2A} and S_{2Anot} are operated as two interleaved pair of switches with 180° phase shift between the two carrier signals as shown in Fig. 2(b).

A. CCM Operation

In the beginning of the switching cycle, when S_{1A} and S_{2Anot} are on, the voltage across the inductor v_L is V_{AN} , hence the inductor current increases linearly. When S_{1A} is turned off at the end of $D_A T_s$ (where T_s is the switching period), S_{1Anot} is turned on and S_{2Anot} is still on, v_L is equal to $V_{AN} - V_{dc}/2$, hence the inductor current decreases. At the end of $T_s/2$, S_{2Anot} is turned off and S_{2A} is turned on, thus v_L becomes equal to V_{AN} again. Similarly, at the end of $T_s/2 + D_A T_s$, S_{2A} is turned off, S_{1Anot} and S_{2Anot} conduct and v_L becomes equal to $V_{AN} - V_{dc}/2$. Thus effectively, the inductor voltage and current frequency (f_l) doubles as compared to the switching frequency (f_s). From inductor volt-sec balance,

$$D_A = 0.5 - V_{AN}/V_{dc} \quad (2)$$

In CCM, the boost inductor L_g is typically selected relatively large to keep the current ripple within 10-40% of peak inductor current. Thus, the current remains either positive or negative for the entire half line cycle and soft-switching is only achieved for the top side devices (S_{1Anot} and S_{2Anot} in Fig. 2(a)).

B. iTCM Operation

In order to achieve soft-switching for S_{1A} and S_{2A} , inductor current needs to be negative like in TCM (as shown in Fig. 2(d)). This duty cycle is varied such that the average current is still sinusoidal, and the switching frequency is varied such that a small negative current, $-I_{ZVS}$ is still achieved during turn-on of both the bottom devices. The value of inductance in TCM is much smaller than CCM such that the current ripple in TCM is approximately twice the peak inductor current. This leads to soft-switching of all devices thus reducing switching losses, however it leads to high ripple on the input current increasing stress on the input EMI filter specifically DM filter. Further, this high ripple boost inductor design very challenging as it needs to be optimized for both the high frequency ripple as well as the low line frequency current. To solve these issues, iTCM is proposed in [5] for single phase two-level converters, in which an LC branch is added between the main inductor and the PWM bridge (as shown in Fig. 1). If $L_g \gg L_b$, this LC branch acts as a high pass filter and most of the high frequency current ripple passes through this branch. The capacitor C_b is added to prevent any low frequency current from passing through L_b . This concept is extended to three-phase three-level converters in this paper where each phase has an LC branch in between the main inductor and the PWM half bridge as shown in Fig. 2(e). Each phase achieves ZVS turn-on while the ripple in the main inductor is small like CCM.

One switching cycle waveform for iTCM for the three-level flying capacitor converter is shown in Fig. 2(f). Each switching cycle can be divided into four intervals. For a balanced symmetrical operation, the flying capacitor must be charged to $V_{dc}/2$, i.e. $V_{Cf} = V_{dc}/2$. C_b acts as a blocking capacitor and blocks the AC voltage. Thus the voltage applied across inductor L_{bA} is given by

$$v_{LbA} = v_{CbA} - v_{bA} \quad (3)$$

Assuming ideal case, i.e. the HF ripple in v_{CbA} is negligible, $v_{CbA} \sim V_{AN}$, thus, effectively paralleling L_{gA} and L_{bA} . The sum of the two currents (i_{LgA} and i_{LbA}) is i_{LegA} which is like the TCM current as shown in Fig. 2(d) and (f), it is divided into L_{gA} and L_{bA} . For $L_{gA} \gg L_{bA}$, most of the HF current flows through L_{bA} as shown in Fig. 2(f). The detailed switching cycle operation is discussed below.

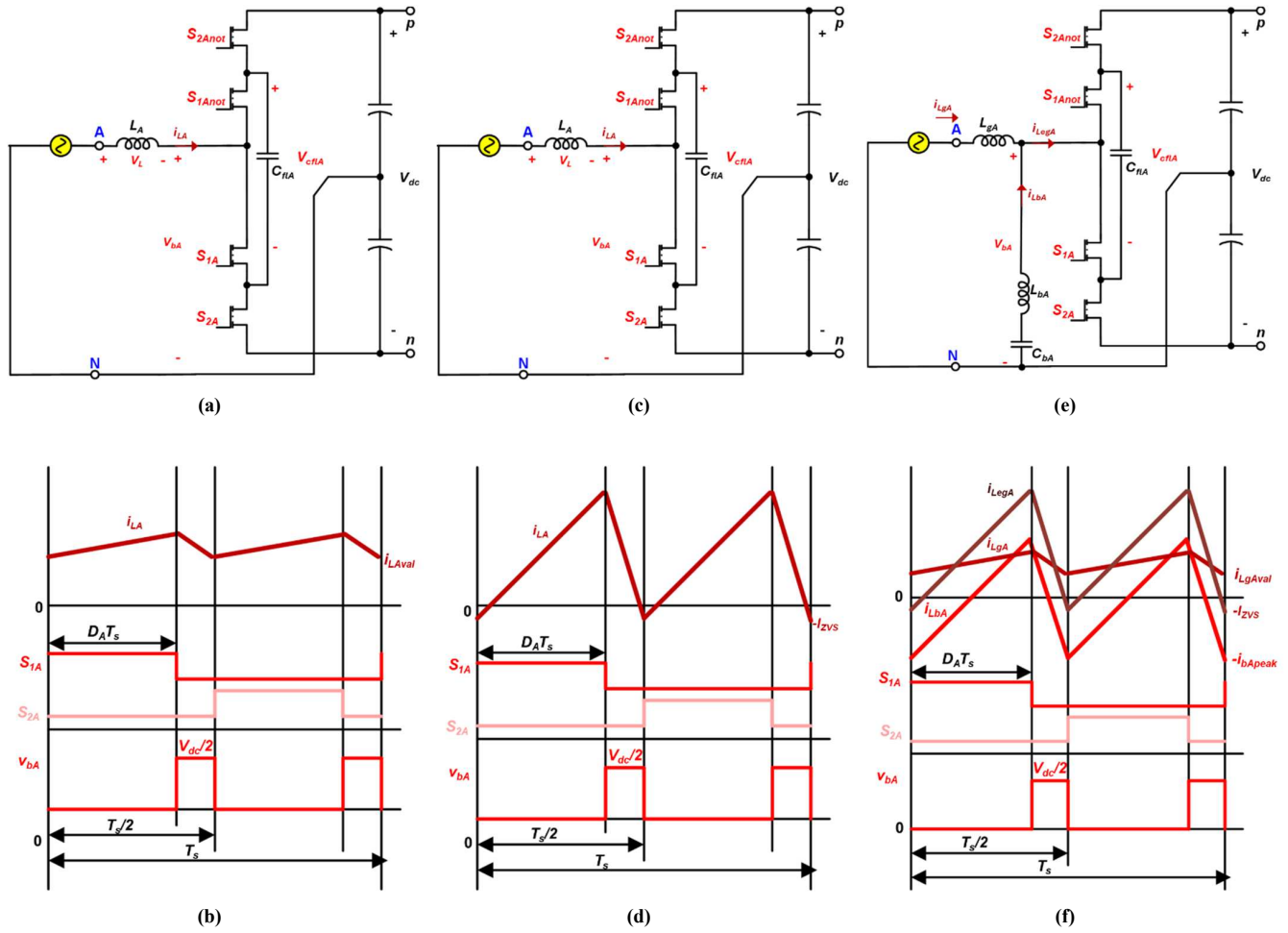


Fig. 2 (a) Equivalent circuit of Phase A during three-phase operation and (b) Gating signals and inductor current during one switching cycle of phase A in CCM mode. (c) Equivalent circuit of Phase A during three-phase operation and (d) Gating signals and inductor current during one switching cycle of phase A in TCM mode. (e) Equivalent circuit of Phase A during three-phase operation and (f) Gating signals and inductor current during one switching cycle of phase A in iTCM mode.

Interval I (0- $D_s T_s$)

In the beginning of the switching cycle, the leg current i_{LegA} starts from a small negative value, $-I_{ZVS}$ as shown in Fig. 2 (f). Since S_{1A} and S_{2Anot} are on, the voltage across the ZVS inductor, L_{bA} is v_{cbA} which is effectively equal to the AC voltage, V_{AN} . Hence i_{Lga} and i_{Lba} increase linearly. S_{1A} is turned off at the end of $D_s T_s$ (where T_s is the switching period). Since the current i_{LegA} is positive, the body diode of S_{1Anot} starts conducting and it is turned on at 0 V.

Interval II ($D_s T_s - T_s/2$)

Once S_{1Anot} is turned on, v_{bA} is equal to $V_{dc}/2$ and v_{Lba} becomes equal to $V_{AN} - V_{dc}/2$, hence i_{Lga} and i_{Lba} decrease linearly. At the end of $T_s/2$, i_{LegA} reaches $-I_{ZVS}$ and S_{2Anot} is turned off. Since the current i_{LegA} is negative, the body diode of S_{2A} starts conducting and it is turned on at 0 V.

Interval III ($T_s/2 - T_s/2 + D_s T_s$)

Once S_{2A} is turned on, v_{Lba} becomes equal to V_{AN} again and i_{Lga} and i_{Lba} increase linearly. At the end of $T_s/2 + D_s T_s$, S_{2A} is turned off and S_{2Anot} is turned on at 0 V.

Interval IV ($T_s/2 + D_s T_s - T_s$)

As S_{1Anot} and S_{2Anot} conduct, v_{bA} becomes equal to $V_{AN} - V_{dc}/2$ and i_{Lga} and i_{Lba} decrease linearly, i_{LegA} falls to zero again. Thus, S_{1A} can be turned on at 0 V.

Thus effectively, the inductor voltage and current frequency (f_L) is doubled as compared to the switching frequency (f_s). Also, ZVS turn-on is achieved for all the devices. The capacitor C_b blocks the AC voltage and hence the $L_b - C_b$ branch acts as a high pass filter with zero low frequency fundamental current. Thus, the fundamental frequency current passes through L_g in addition to a small high frequency ripple.

Active Balancing of Flying Capacitor Voltage

For symmetric operation and equal duty ratios for inner and outer pair of devices, flying capacitors are naturally balanced at $V_{DC}/2$. However practically due to asymmetric delays in gate signals or asymmetric gate loop and power loop design, the flying capacitor voltage can deviate from $V_{DC}/2$ which can lead to higher voltage stress on one set of devices and loss of frequency multiplication effect. Hence for robust operation a feedback controller is added [6]. The feedback control is

designed based on [6] with a proportional controller adjusting the duty ratio based on the error in V_{Cfl} .

III. PARAMETERS DESIGN

The boost inductor L_g is designed in such a way that it carries only a small current ripple $\Delta i_{Lg,pp}$, while the current ripple in L_b has to be chosen, such that i_{LegA} reaches a negative value of at least $-I_{ZVS}$ in the corresponding switching transitions, in order to guarantee soft-switching over the whole mains period. Thereby, the current I_{ZVS} is the desired turn-off current of the HF leg, which can be calculated based on the effective output charge (Q_{oss}) of the switching devices required to charge/discharge the output capacitances (C_{oss}) of the GaN HEMTs, and the maximum allowed duration of this resonant switching transition (i.e., the dead time duration T_{dt}) [5]. The dead time for the turn-on transition ($T_{dvalley}$) when the current is at its valley point ($-I_{ZVS}$) has to be small percentage of the switching period. The minimum required current I_{ZVS} can roughly be calculated as

$$I_{ZVS} = Q_{oss} / T_{dvalley} \quad (4)$$

The critical point, where the highest current in L_b is needed, is at the peak of the mains current. The inductance values of L_b and L_g is designed in such a way that a turn-off current of $-I_{ZVS}$ is achieved in this operating point. With the known minimum switching frequency and the purpose of achieving a turn-off current of $-I_{ZVS}$ at the peak of the mains current, the inductance of L_b can be calculated as a function of the peak-to-peak current ripple r (in % of the peak mains current) in the boost inductor L_g as:

$$L_b = \frac{2V_{phrms} \left(\frac{V_{dc}}{2} - V_{phmax} \right) \left(\frac{2V_{phmax}}{V_{dc}} \right) \frac{2}{f_{Lmin}}}{2I_{ZVS}V_{phrms} + \frac{\sqrt{2}P}{3}(2-r)} \quad (5)$$

where V_{phmax} denotes the peak AC grid voltage, V_{dc} denotes the DC-link voltage, P denotes the system power, and f_{Lmin} is the desired minimum inductor current frequency. Furthermore, the inductance of the boost inductor L_g is designed for a maximum current ripple r , can be found as

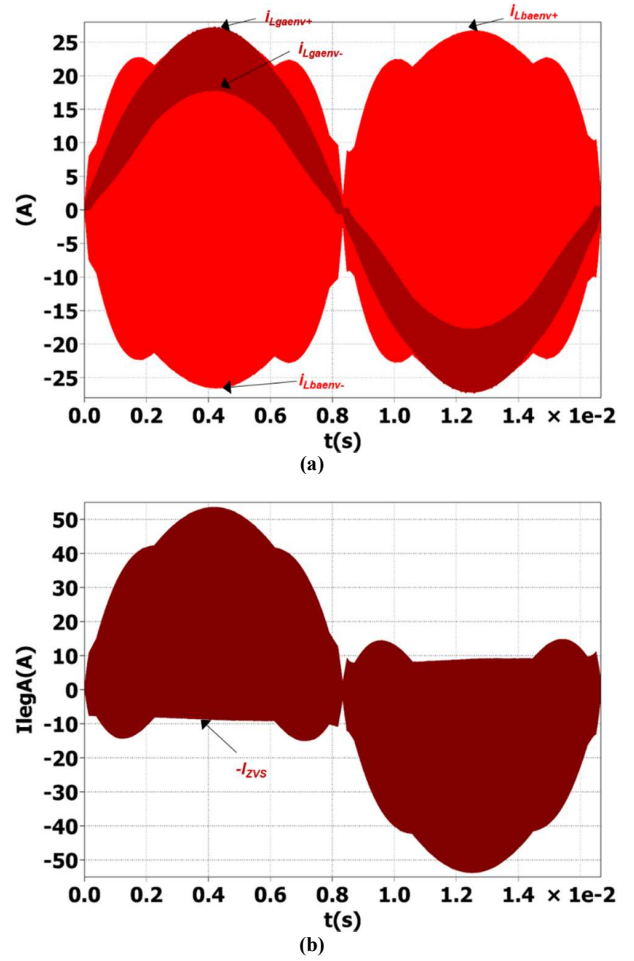
$$L_g = \left(1 - \frac{V_{phmax}}{V_{dc}/2} \right) \frac{2V_{phrms}^2}{f_{Lmin} r P / 3} \quad (6)$$

The simulated waveforms for a ripple, $r=40\%$ and system specifications from Table I are shown in Fig. 3.

The capacitance of the ZVS branch C_b must be dimensioned for a maximum peak-to-peak voltage ripple r_v (in % of the peak mains voltage):

$$C_b = \frac{2i_{baenv+p}(0.5 - V_{phmax}/V_{dc})}{r_v V_{phmax} f_{Lmin} / 2} \quad (7)$$

where $i_{baenv+p}$ is the peak of upper envelope $i_{ba,env+}$ of the current in L_b (shown in Fig. 3(a)). There is a tradeoff between maximum voltage ripple allowed and phase shift caused by C_b in between



mains voltage and AC current. The simulated waveforms shown in Fig. 3 are for $r_v = 2.5\%$. The switching frequency is varied such that a constant negative current, $-I_{ZVS}$ is achieved for ZVS

Fig. 3(a) Simulated phase A currents in L_g and L_b showing envelope of currents for both inductors for specifications shown in Table I. (b) Leg current I_{legA} achieving ZVS current $-I_{ZVS}$ for soft-turn-on of bottom devices.

turn-on for both the bottom devices during positive half line cycle.

TABLE I. CONVERTER SPECIFICATIONS

Parameter	Value	Parameter	Value
V_{phrms}	230 V	f_{iL}	380-760 kHz
V_{dc}	800 V	C_{DC}	2*1.56 mF
P	11 kW	I_{ZVS}	8 A
f_{sw}	190-380 kHz	$T_{dvalley}$	120 ns

Switching Frequency Calculation

The condition of a constant ZVS current I_{ZVS} can be visualized, e.g., during the positive half cycle of the mains period as the difference of the upper envelope $i_{ba,env+}$ of the current in L_b and the lower envelope $i_{ga,env-}$ of the current in L_g ,

as shown in Fig. 3(a). The function to describe the variable inductor current frequency (f_L) to achieve a constant ZVS current can be found as

$$f_L = A_{ta} \frac{1/L_{eq}}{2(|I_{acaref}| + I_{ZVS})V_{phmax}} \quad (8)$$

$$L_{eq} = L_g \parallel L_b \quad (9)$$

$$A_{ta} = |\sin(\omega t)| (1 - |\sin(\omega t)|) \frac{V_{phmax}}{V_{dc}/2} \quad (10)$$

Here, the effect of C_b and any filter capacitances is neglected and it is assumed that i_{Lg} will be perfectly in phase with the grid voltage. This is a reasonable assumption as for the specifications shown in Table I, $f_{Lmin} \gg f_{line}$ and the phase-shift caused by C_b at line frequency is $<5\%$.

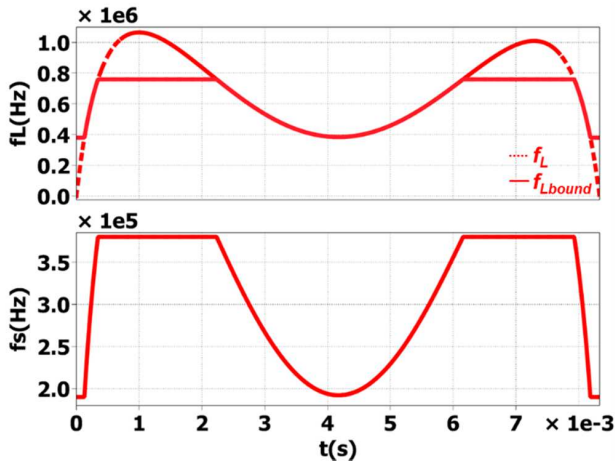


Fig. 4. Inductor current frequency and switching frequency variation to maintain constant negative current I_{ZVS} during turn-on of both bottom devices with limit on maximum and minimum switching frequency.

Flying Capacitor Design

The flying capacitor C_{fl} is dimensioned in such a way that it carries only a small voltage ripple Δv_{Cflpp} , on top of the average of the flying capacitor voltage, V_{CFLavg} . V_{CFLavg} and Δv_{Cflpp} are given by:

$$V_{CFLavg} = \frac{V_{DC}}{N-1} \quad (11)$$

$$\Delta v_{CFLpp} = \frac{2PV_{dc}}{3V_{phmax}^2 C_{fl} f_s} D \left(\frac{1}{2} - D \right) \quad (12)$$

Where N is the number of levels, and D is given by (2). For three-level flying capacitor converter, $V_{CFLavg} = V_{DC}/2$. The peak-peak ripple is maximum when AC current ripple is maximum which in case of unity power factor, is also when AC voltage is maximum. Since the switching frequency is also minimum at this point (from Fig. 4), hence C_{FL} has to be dimensioned such that this ripple is a small percentage (r_{cfl}) of V_{CFLavg} , which is given by:

$$C_{fl} = \frac{4PV_{dc}}{3V_{phmax}^2 r_{cfl}(V_{dc}) f_{smin}} \left(\frac{1}{2} - \frac{2V_{phmax}}{V_{dc}} \right) \left(\frac{2V_{phmax}}{V_{dc}} \right) \quad (13)$$

For $r_{cflmax}=0.6\%$, $C_{fl}=6.58 \mu F$. The parameters values are summarized in Table II, the variation of Δv_{Cflpp} over half line cycle is shown in Fig. 5 for the converter specifications in Table I and parameters in Table II.

TABLE II. PARAMETERS VALUES

Parameter	Value	Parameter	Value
L_g	16.7 μH	C_b	2 μF
L_b	3 μH	C_{fl}	6.58 μF

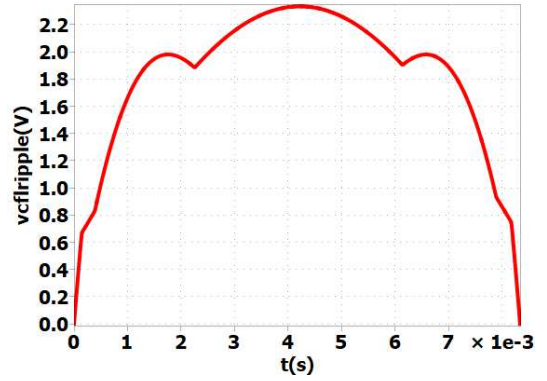


Fig. 5. Flying capacitor voltage ripple variation over half line cycle for converter specifications and parameters in Table I&II respectively.

IV. HARDWARE DESIGN

The devices used in the high frequency phase legs are 2*GS66516T (blocking voltage: 650 V) in parallel. The total value of the flying cap is 6.58 μF , the combination is realized by 14*0.47 μF ceramic capacitors. The GaN based flying capacitors phase leg design and inductors' design is discussed in detail below.

A. GaN based Flying Capacitor Phase Leg Design

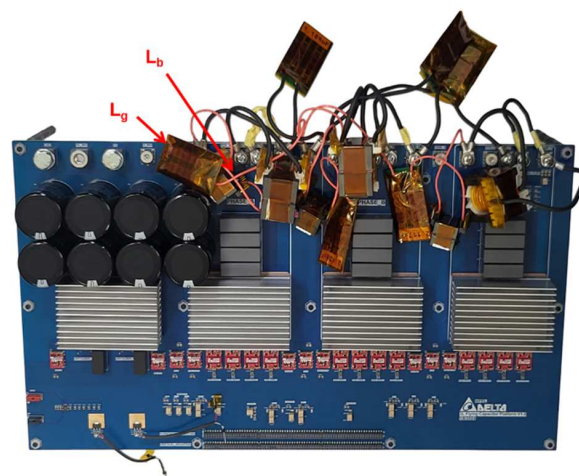
The phase leg design is based on [6], owing to the fast transients in GaN devices, it is necessary to design small gate loops and power loops to avoid any overvoltage on the gate as well as the drain. It is also very important to design the gate loop and power loop completely symmetrical for the two GaN devices in parallel to prevent any current mismatch between the devices as well as any oscillations on the gate voltage during transient. Secondly, it is of utmost importance that the two sets of complementary devices have symmetrical gate loop and power loop as a mismatch in these critical loops design will cause an unbalance in the flying capacitor voltage. The small power loop design with decoupling caps right next to the devices is shown in Fig. 6. The top layer (shown in Fig. 6(a)) has the decoupling caps for the outer pair of devices while the bottom layer has the decoupling caps for the inner pair of complementary devices (as shown in Fig. 6(b)). The gate drivers are isolated comprising of two stages: an isolator + gate driver. The non-isolated gate drivers are placed on the bottom layer

B. Inductor Design

V. EXPERIMENTAL VERIFICATION

TABLE III. COMPONENTS

Component	Description
GaN devices	2*GS66516T (650 V, 60 A) in parallel/ switch
Slow leg devcies for 1-ph operation	2*NTH4L014N120M3P (1.2 kV, 127 A) in parallel/ switch
Flying capacitors	14*C2220C474KCRCTU (MLCC - SMD/SMT 500V 0.47uF X7R 2220)/ phase
DC bus capacitors	8*ALA7DA391DF550 (Aluminum Electrolytic Capacitors - 550V 390 uF)



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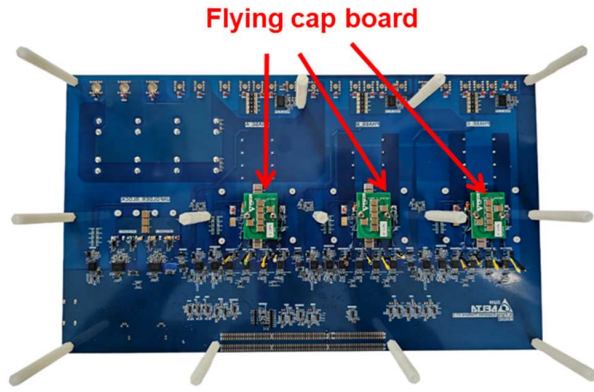
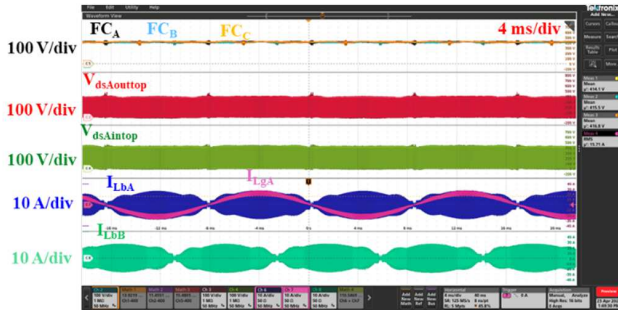
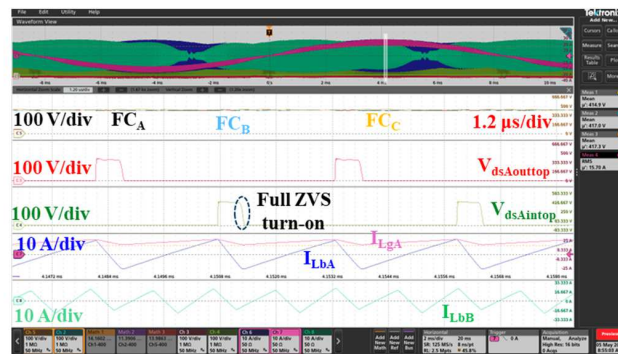


Fig. 10. Bottom side of the board showing flying capacitor daughter card

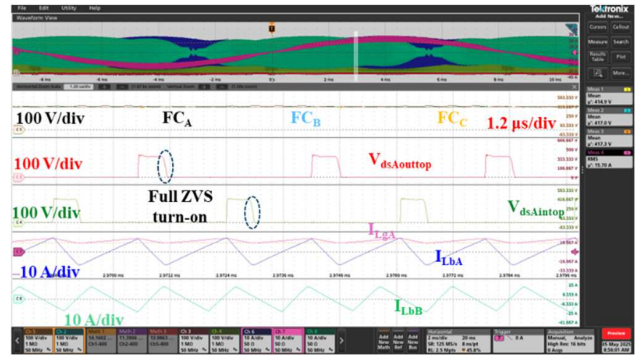
The key experimental waveforms for converter specifications in Table I are shown in Fig. 11. The flying capacitor voltage is very close to 400 V (as shown in the figure) for all the three phases, thus achieving good balancing. The measured flying capacitor voltage for phase A, B and C is 404.7 V, 400.8 and 404.7 V respectively. Full ZVS turn-on is achieved with $I_{ZVS} = -8$ A as shown in Fig. 11(b) and (c). The frequency multiplication effect is also shown in Fig. 11(b) and (c). The measured efficiency is 98.35 %.



(a)



(b)



(c)

Fig. 11. Key experimental waveforms at $V_{phrms} = 230$ V, $V_{dc} = 800$ V, $P_o = 11$ kW, $f_s = 190$ -380 kHz (a) Multiple line cycles, (b) & (c) Zoom-in waveforms showing full ZVS turn-on at $\phi = 90^\circ$ and 60° respectively. From top to bottom: 1. Flying capacitor voltages for phase A, B and C 2. $V_{dsAouttop}$ 3. $V_{dsAintop}$ 4. Phase A I_{Lg} and I_{Lb} currents 5. Phase B I_{Lb} current.

Efficiency vs I_{ZVS} and Dead Time

The effect of varying I_{ZVS} and peak dead time is shown in Fig. 12. The efficiency increases by 0.45% when reducing the dead time from 150 ns to 80 ns. The measured efficiency vs load for $V_{phrms} = 230$ V, $V_{dc} = 800$ V, $f_{iL} = 380$ -760 kHz, $f_s = 190$ -380 kHz is shown in Fig. 13, the peak efficiency achieved is 98.35% at $P_o = 11$ kW. The loss breakdown based on a model for full load is shown in Fig. 14. The detailed component losses are shown in Table IV. The switching losses are calculated based on the double pulse test (DPT) data from [10]. The calculated efficiency is 98.76% which is close to the measured efficiency. The losses in L_b are based on ANSYS Maxwell simulation model. The core losses are more dominant than winding losses, which can be improved by employing custom core designs such that the effective volume of the core can be reduced while maintaining similar effective area. Further, distributed air gap should help in reducing the losses caused by fringing effect [11].

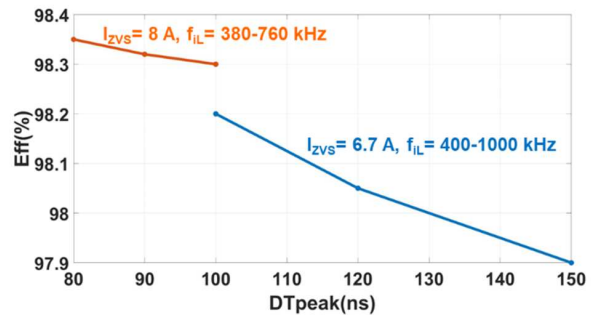


Fig. 12. Efficiency vs dead time and I_{ZVS} for $V_{phrms} = 230$ V, $V_{dc} = 800$ V, $P_o = 11$ kW, $f_s = 190$ -380 kHz

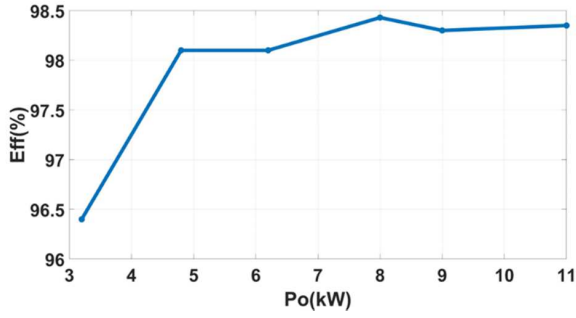


Fig. 13. Measured efficiency vs load at $V_{phrms}= 230$ V, $V_{dc}= 800$ V, $f_{il}= 380$ -760 kHz, $f_s=190$ -380 kHz

TABLE IV. LOSS BREAKDOWN BASED ON CALCULATION

Component	Loss (W)
GaN conduction (P_{cond})	52.05 ($T_j= 65^\circ\text{C}$)
GaN turn-off (P_{sw})	3.7
L_b core loss (P_{clb})	18.3
L_b winding loss (P_{wlb})	6.36
L_g core loss (P_{clg})	1.2
L_g winding loss (P_{wlg})	6.41
GaN dead-time conduction (P_{cond_DT})	39.8
Total	136.3
Projected Efficiency (%)	98.76

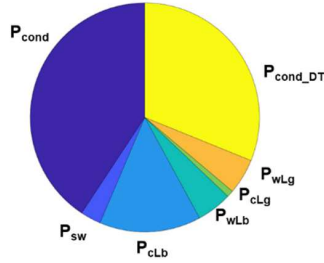


Fig. 14. Loss breakdown based on loss model at $V_{phrms}= 230$ V, $V_{dc}= 800$ V, $P = 11$ kW, $f_{il}= 380$ -760 kHz, $f_s=190$ -380 kHz.

Efficiency Improvement with Lower DC Bus Voltage

It can be inferred from the loss breakdown that the most dominant losses are the device conduction losses which is an expected characteristic of TCM/ iTCM converters. However, deadtime conduction losses and L_b core losses are also significant. Deadtime conduction losses can be reduced by reducing dead time and employing variable dead time over AC line cycle. The core losses in L_b can be reduced by improving inductor design and employing custom core shapes as discussed in previous section. The losses can also be reduced by reducing switching frequency which can be achieved in iTCM by reducing DC bus voltage. The efficiency load curve for $V_{dc}= 750$ V is shown in Fig. 15. The efficiency achieved at $V_{phrms}= 230$ V, $V_{dc}= 750$ V, $f_{il}= 300$ -760 kHz, $f_s=150$ -380 kHz with the same set of passive components (as in Table II) is 98.76%.

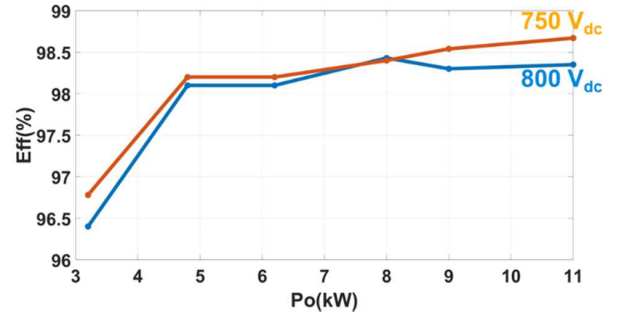


Fig. 15. Measured efficiency vs load at $V_{phrms}= 230$ V, $f_{il}= 300$ -760 kHz, $f_s= 150$ -380 kHz

VI. CONCLUSION

In this paper, a three-phase GaN based flying capacitor three-level converter based on the iTCM concept is demonstrated at 800 V DC bus voltage and 11 kW output power. The iTCM concept is a simple method to achieve soft switching over the entire AC grid period by adding an LC circuit to the well-known full-bridge PWM AC/DC converter. It is shown that by the addition of just one LC branch, soft-switching can be achieved for all the devices while still filtering the large high frequency ripple in TCM. It is demonstrated that a constant ZVS current and, therewith, soft switching over the whole grid period can be achieved even in open-loop operation, i.e., no current zero crossing detection, as in case of TCM, is required. The switching operation is discussed in detail showing that the inductor current frequency is double the switching frequency. A robust active flying capacitor voltage balancing method based on flying capacitor voltage measurement is also presented. The design of all main power components and passive components of the converter is presented. The three-level GaN phase leg design is also discussed in detail stating the importance of small gate loop and power loop and symmetry between the two sets of complementary switches to achieve good balancing for flying capacitor voltage. Experimental results are shown for DC bus voltage of 750-800 V. The measured efficiency is 98.76% at $V_{phrms}= 230$ V, $V_{dc}= 750$ V, $P= 11$ kW, $f_s=150$ -380 kHz, $f_{il}= 300$ -760 kHz with SPWM.

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